

InterPro Classification of protein families

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Overview Entries 14 Sequence

Sequence ID Norm.seq

Sequence Length 1068 amino acids

Protein family membership **Phosphatidylinositol 3-/4-kinase** (IPR015433)

Entry matches to this protein¹

Feature Display Mode Summary Full

Families

Domains

Representative families

Representative domains

Phosphoinositide 3-kinase family, accessory domain (PIK domain)
Integrated: IPR001263
520 - 703

Q3: I inputted the protein sequence into PFAM/InterPro sequence search, which outputted 5 PFAM domain matches (“blue bars” in top screenshot, and listed out in bottom screenshot). The Q2 mutations cluster specifically to the PI3Ka domain (residues 520-703), PF00613.

InterPro Classification of protein families

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Result / InterProScan / Iprscan5-R20260609-015315-0620-22171566-P1m / Entry / Pfam

InterProScan Search Result **InterProScan protein**

Overview Entries 14 Sequence

This protein matches these entries:

1 - 5 of 5 entries matching Pfam

Accession	Short Name	Name	Source Database	Matches
PF02192	PI3K_p85B	PI3-kinase family, p85-binding domain	Pfam	500 - 1000
PF00613	PI3Ka	Phosphoinositide 3-kinase family, accessory domain (PIK domain)	Pfam	500 - 1000
PF00794	PI3K_rbd	PI3-kinase family, ras-binding domain	Pfam	500 - 1000
PF00792	PI3K_C2	Phosphoinositide 3-kinase C2	Pfam	500 - 1000
PF00454	PI3_P14_kinase	Phosphatidylinositol 3- and 4-kinase	Pfam	500 - 1000

Rows per page: 20

Previous 1 Next

Q3: Same output as above (5 PFAM matches, with PI3Ka specific to mutations), using HMMER search instead

Browse Phmmer Results

[Score](#) [Taxonomy](#) [Domain](#) [Download](#)

Go to Batch job summary

PI3Ka (PF00613)

Description: Phosphoinositide 3-kinase family, accessory domain (PIK domain)

Coordinates: 519 - 704 (alignment region 520 - 703)

Model Match: 1 185

Source: pfam

PI3K_rbd

PI3K_C2

PI3Ka

PI3_P14_kinase

Distribution of Significant Hits

Bacteria

Archaea

Eukaryota

Viruses

Unclassified Sequences

Other Sequences

more significant

Search Details

Customise Table

Target	Description	Species	E-value
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Q4: Top 10 observed mutations (all missense substitutions) in PIK3CA-encoded protein.

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Q

e.g. BRAF, Breast, TCGA-BLCA, TCGA-A5-A0G

GENE

PIK3CA

MOST FREQUENT SOMATIC MUTATIONS

Save/Edit Mutation Set

TSV

TOTAL OF 436 SOMATIC MUTATIONS

Q

Search

☐	DNA Change	Protein Change	Type	Consequences	# Affected Cases in PIK3CA	# Affected Cases Across the GDC	Impact
☐	chr3:g.179218303G>A	PIK3CA E545K	Substitution	Missense	+ 332 / 5,791 (5.73%)	332 / 18,289 (1.82%)	MO DH PR
☐	chr3:g.179234297A>G	PIK3CA H1047R	Substitution	Missense	+ 303 / 5,791 (5.23%)	303 / 18,289 (1.66%)	MO TO BE
☐	chr3:g.179218294G>A	PIK3CA E542K	Substitution	Missense	+ 204 / 5,791 (3.52%)	204 / 18,289 (1.12%)	MO DH PR
☐	chr3:g.179199088G>A	PIK3CA R88Q	Substitution	Missense	+ 85 / 5,791 (1.47%)	85 / 18,289 (0.46%)	MO TO PR
☐	chr3:g.179234297A>T	PIK3CA H1047L	Substitution	Missense	+ 49 / 5,791 (0.85%)	49 / 18,289 (0.27%)	MO TO BE
☐	chr3:g.179203765T>A	PIK3CA N345K	Substitution	Missense	+ 41 / 5,791 (0.71%)	41 / 18,289 (0.22%)	MO DH PR
☐	chr3:g.179221146G>A	PIK3CA E726K	Substitution	Missense	+ 37 / 5,791 (0.64%)	37 / 18,289 (0.20%)	MO TO BE
☐	chr3:g.179210192T>C	PIK3CA C420R	Substitution	Missense	+ 34 / 5,791 (0.59%)	34 / 18,289 (0.19%)	MO TO PO
☐	chr3:g.179199690G>A	PIK3CA G118D	Substitution	Missense	+ 30 / 5,791 (0.52%)	30 / 18,289 (0.16%)	MO TO PO
☐	chr3:g.179210291G>A	PIK3CA E453K	Substitution	Missense	+ 25 / 5,791 (0.43%)	25 / 18,289 (0.14%)	MO TO PO

Show

10

Entries

Showing 1 - 10 of 436 somatic mutations

<<

<

1

2

3

4

5

...

44

>

>>

Q5: Top 10 TCGA projects affected by mutations of PIK3CA gene

portal.gdc.cancer.gov

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Note: Projects are, by default, ranked by percentage of cases affected by PIK3CA mutations.

GENE • PIK3CA

JSON TSV

TOTAL OF 44 PROJECTS

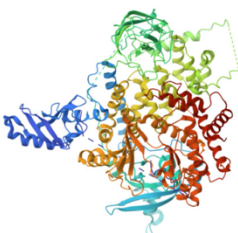
Project	Disease Type	Primary Site	# SSM Affected Cases	# CNV Amplifications	# CNV Homozygous Deletions	# Mutations
TCGA-UCEC	4 Disease Types	2 Primary Sites	253 / 512 (49.41%)	51 / 540 (9.44%)	0 / 540 (0.00%)	151
TCGA-UCS	2 Disease Types	2 Primary Sites	20 / 57 (35.09%)	15 / 55 (27.27%)	0 / 55 (0.00%)	15
TCGA-BRCA	9 Disease Types	Breast	331 / 969 (34.16%)	131 / 1,084 (12.08%)	0 / 1,084 (0.00%)	76
CML-MBC	Ductal and Lobular Neoplasms	Breast	52 / 174 (29.89%)	0 / 0 (0.00%)	0 / 0 (0.00%)	17
TCGA-COAD	4 Disease Types	2 Primary Sites	125 / 428 (29.21%)	4 / 454 (0.88%)	0 / 454 (0.00%)	59
CGIC-HTMCP-CC	4 Disease Types	Cervix uteri	24 / 85 (28.24%)	44 / 122 (36.07%)	0 / 122 (0.00%)	11
TCGA-CESC	4 Disease Types	2 Primary Sites	81 / 287 (28.22%)	124 / 299 (41.47%)	0 / 299 (0.00%)	31
TCGA-BLCA	4 Disease Types	Bladder	86 / 408 (21.08%)	50 / 402 (12.44%)	0 / 402 (0.00%)	42
CPTAC-2	5 Disease Types	6 Primary Sites	64 / 328 (19.51%)	0 / 0 (0.00%)	0 / 0 (0.00%)	40
TCGA-HNSC	Squamous Cell Neoplasms	13 Primary Sites	86 / 509 (16.90%)	176 / 519 (33.91%)	0 / 519 (0.00%)	35

Showing 1 - 10 of 44 projects

Q6: PDB ID 2RD0

Download Multiple Files: All Selected

Download File View File Download Alignment



Explore in 3D

2RD0: Entity 1

Structure of a human p110alpha/p85alpha complex

Huang, C., Gabelli, S.B., Amzel, L.M.

(2007) Science 318: 1744-1748

Released2007-12-25

MethodX-RAY DIFFRACTION 3.05 Å

Chain IDA

OrganismHomo sapiens

MacromoleculePhosphatidylinositol-4,5-bisphosphate 3-kinase catalytic subunit alpha isoform

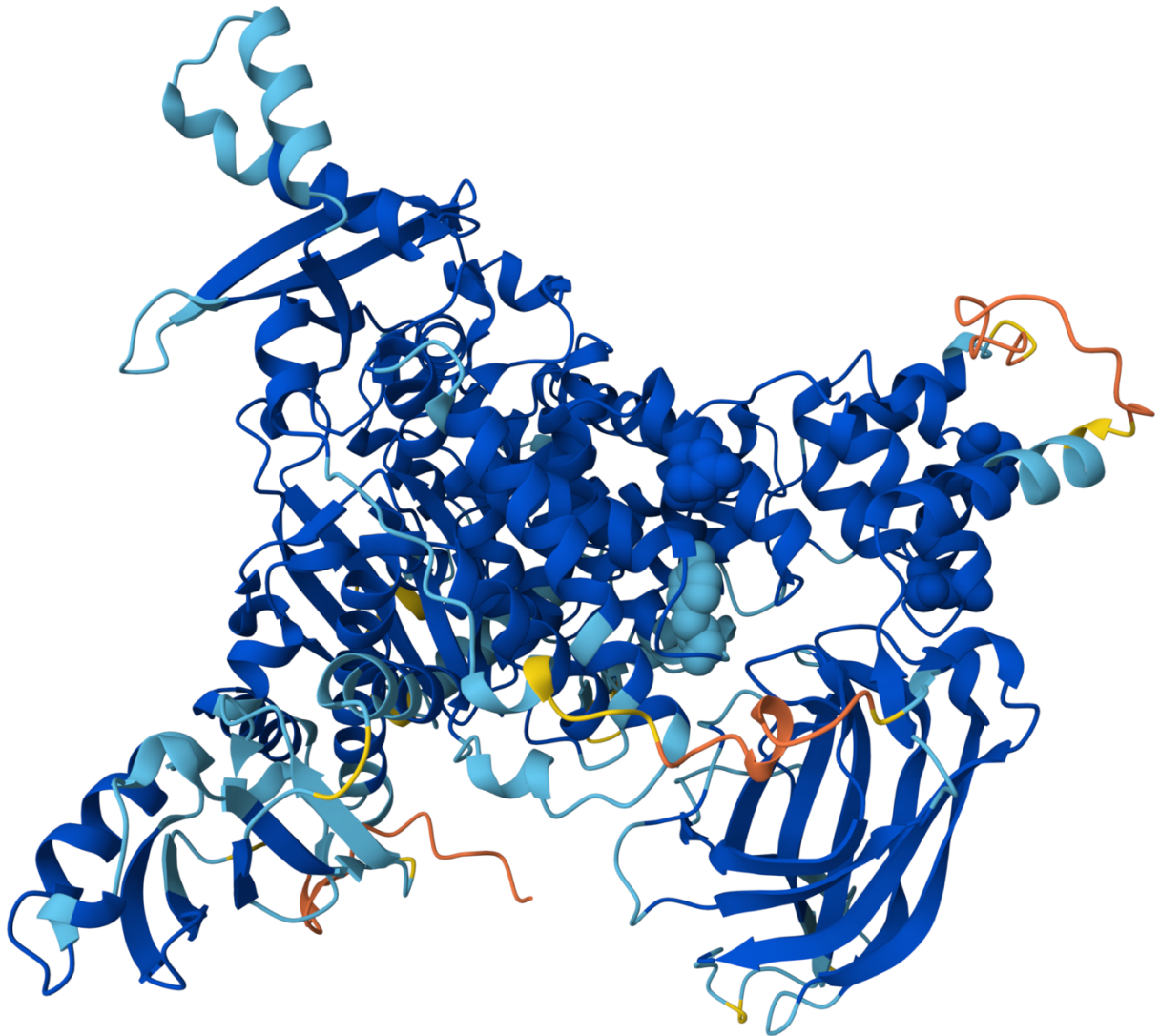
Sequence MatchSequence Identity: 100%, E-Value: 0, Region: 29-1096

01002003004005006007008009001,000

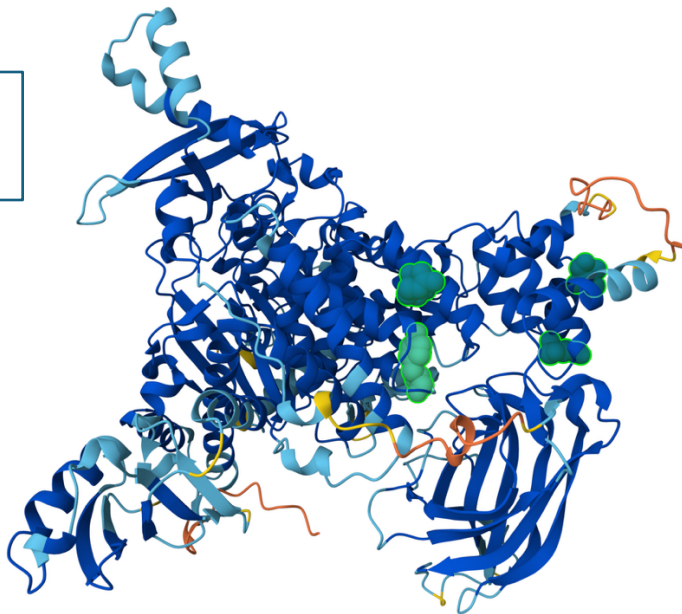
QUERY2RD0_1

Note: 2RD0 is the PI3Kalpha dimer, with (1) the PIK3CA-encoded p110alpha catalytic subunit (2RD0_1; shown above) and (2) the p85alpha regulatory subunit (2RD0_2). The wt_healthy query sequence for PIK3CA-encoded p110alpha is covered by residues 29-1096 of 2RD0_1.

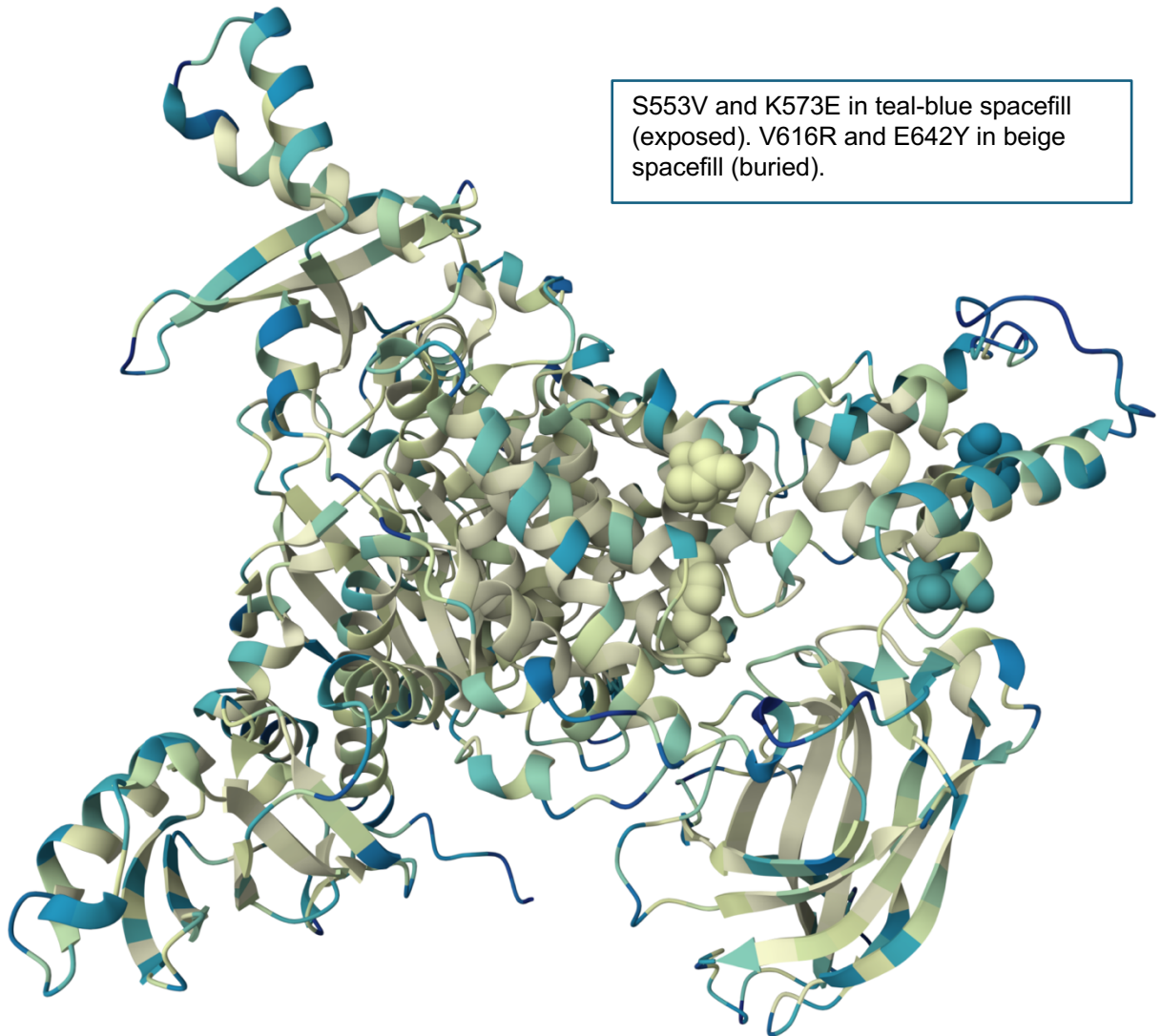
Q7: AlphaFold structural model of mutant sequence, colored by pLDDT quality score. Mutant amino acid side chains are highlighted as spacefill.



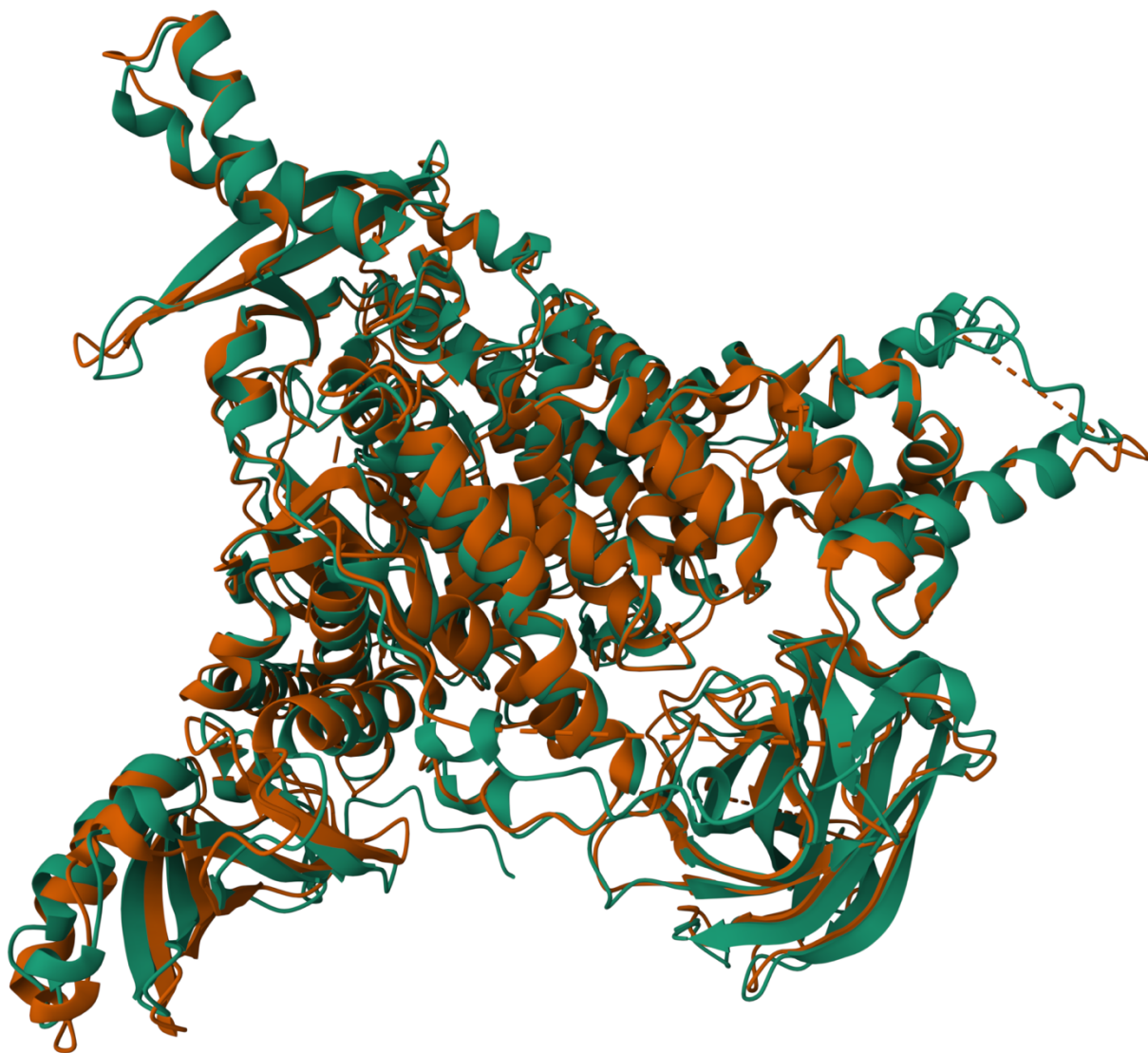
Another image, with
residues selected for
easier locating.



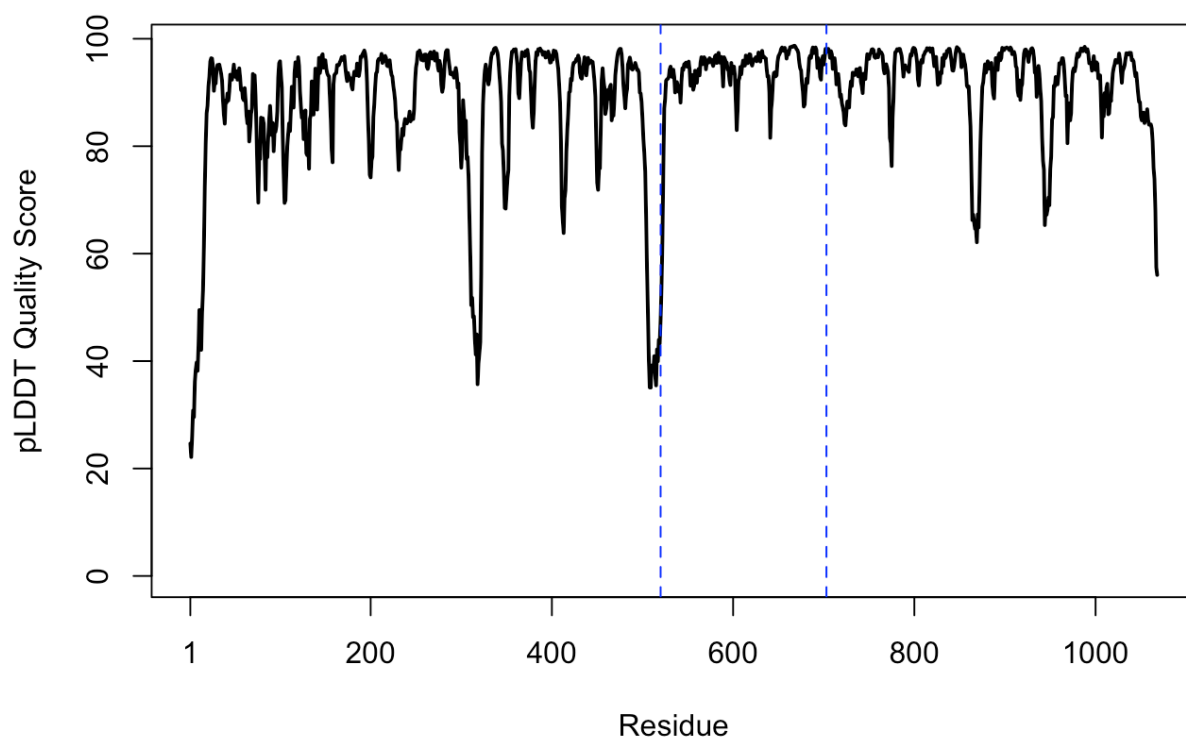
Q8: Same structure shown above, now colored by “Accessible Surface Area” (beige corresponds to buried residues, whereas blue corresponds to exposed residues)



Q9: Superposition of AlphaFold structure prediction (green) and PDB 2RD0_1 (orange).



Plot of pLDDT Values for AlphaFold Rank 001 Model



PAE Score Plot for AlphaFold Rank 001 Model

